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A Simple Structure Constrained Attitude Control for Rigid Bodies: A PD-Type Control

MEHDI GOLESTANI[1], KHALID A. ALATTAS[2], SAMI UD DIN[3], (Member, IEEE), SALEH MOBAYEN[1], (Senior Member, IEEE), ABDULLAH K. ALANAZI[4], AND AFEF FEKIH[5], (Senior Member, IEEE)

[1]Future Technology Research Center, National Yunlin University of Science and Technology, Douliou, Yunlin 64002, Taiwan
[2]Department of Computer Science and Artificial Intelligence, College of Computer Science and Engineering, University of Jeddah, Jeddah 23890, Saudi Arabia
[3]Department of Electrical Engineering, Namal University Mianwali, Mianwali 42250, Pakistan
[4]Department of Chemistry, Faculty of Science, Taif University, Taif 21944, Saudi Arabia
[5]Department of Electrical and Computer Engineering, University of Louisiana at Lafayette, Lafayette, LA 70504, USA

Corresponding author: Saleh Mobayen (mobayens@yuntech.edu.tw)

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ABSTRACT This study investigates the challenging and complicated issue of full-state constraint attitude control for rigid bodies under actuators physical limitation. Using the concept of prescribed performance control (PPC), a novel proportional-derivative (PD)-type control is introduced by which both the attitude quaternion and the rotation velocity of the rigid body are enforced to possess specific behaviors in transient and steady state. To this end, a prescribed performance function (PPF) with finite-time convergence is first defined as the predefined boundaries for the quaternion and rotation velocity. Subsequently, a constrained PD-type attitude control for rigid body attitude system is developed. The significant difference between the proposed methodology and the PPC is the simple structure of the controller making it more applicable from practical implementation point of view. Indeed, due to the use of error transformation in the PPC, the controller contains partial derivative terms and complicated functions even for only constraining the attitude quaternion. When it comes to angular velocity constraint as well, the complexity of the control design procedure is doubled. It is rigorously proved that the suggested control framework can successfully satisfy constraints not only on the quaternion but also on the angular velocity, simultaneously. These interesting results are obtained even when the actuator saturation is considered. The simulation results conducted on a rigid spacecraft verify the efficacy and applicability of the suggested constrained attitude control approach.

INDEX TERMS Attitude control, rigid body, full-state constraint, input saturation, PD-type control.

I. INTRODUCTION

Attitude control system, as an important part of rigid bodies such as spacecraft [1], [2] and UAVs [3], [4], plays a crucial role in missions success. Therefore, it has attracted attention of researchers and numerous control strategies have been developed to obtain desirable attitude performance. For instance, backstepping method [5], variable structure control [6], event-triggered control [7], model predictive control (MPC) [8], adaptive control [9], H_∞ control [10], fuzzy control [11], output feedback control [12] and disturbance observer-based control [13] were presented to develop attitude control for rigid bodies. Although the attitude control problem has witnessed a considerable development, there is still a challenging issue on how to design an effective attitude control in spite of the actuators saturation and external disturbance. Considering these two factors, a wide variety of attitude controls for rigid body systems has been presented; see for example [14]–[17].

Despite the fact that there are numerous methods to design an attitude control, the PD control is still recognized as the most common and practical approach because of its simple structure, independency of modelling parameter, low computational effort and explicit tuning procedures. A saturated PD control for spacecraft has been presented in [18] such that there is no need for the angular velocity to be measured. In [19], based on the passivity concept, a PD control for spacecraft attitude control problem has

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been proposed. A hybrid PD control based on input-to-state stability (ISS) for a rigid spacecraft subject to external disturbance has been given in [20]. The PD+ approach which is composed of a linear PD control and a nonlinear dynamic inversion method is used to remove the model's nonlinear terms based on feedback linearization concept. The research related to PD+ technique for spacecraft attitude control can be found in [21] and the references therein. Despite the efficacy and appropriate performance of the suggested PD control, it only guarantees that the system state trajectories are drawn into a small vicinity of the origin. In addition, these results can only achieve the steady-state performance of the rigid body attitude system, while the transient-state performance is also a major concern in the attitude maneuvering.

The pioneering research in providing predefined performance in transient and steady state for nonlinear systems was carried out in [22] and then followed in many works such as [23]–[29]. This concept has also been employed in attitude control design for rigid bodies. More specifically, A constrained terminal sliding mode attitude control for quadrotor considering actuator saturation was introduced in [30]. To enhance the efficacy of the satellite and to prevent damages to the satellite due to poor tracking performance, a fault-tolerant attitude control with guaranteed performance has been presented in [31]. In [32], a PPC for spacecraft attitude stabilization and tracking has been proposed in which the rotation velocity of the spacecraft is estimated with a differentiator. The problem of PPC for trajectory tracking on SO(3) has been studied in [33]. A new learning-based PPC for spacecraft formation subject to external disturbance has been proposed in [34]. Inspired by the PPC notion, a constrained attitude tracking for spacecraft has been presented in [35] such that the maximum convergence time of the closed-loop system can be pre-determined by designer. Using backstepping approach, Ref. [36] designed a fault-tolerant PPC for rigid spacecraft so that the convergence speed of the attitude quaternion can be defined *a priori*.

Despite the fact that the PD-type control problem for rigid body attitude stabilization has been studied in the past, a constrained PD-type control approach is proposed in this paper which is able to provide the prescribed performance in transient and steady-state for full states. However, the existing PD or PD-type controls cannot guarantee such performance. Moreover, the existing constrained attitude control schemes [33]–38] for constraining only the attitude variable (not full states) generally use the concept of the transformation error presented in [22]. The key idea is to provide an error transformation that transforms the original "constrained" system into an equivalent "unconstrained" one. It is proven that stabilizing the "unconstrained" system is sufficient to achieve the guaranteed prescribed performance. A controller is then designed to guarantee a uniform ultimate boundedness property for the transformed error. However, the aforementioned constraint methods and controller structures are complex; because they fully use the transformed error and they involve considerable intricate function terms as well as partial differential terms even when the attitude variable constraint is only considered.

Indeed, in order to improve pointing accuracy and pointing stability, both attitude variable and angular velocity should be constrained simultaneously. Thus, constraining all the system states based on this concept considerably increases complexity of the control structure. Instead of fully use of the transformed attitude variable in the controller design procedure, our proposed constrained control is composed of two parts: The first part is a conventional PD control which is used to make the system states converge to zero. The second one is an auxiliary control to preclude the system trajectories from intersecting the prescribed performance functions. Hence, to guarantee the predefined control performance for full states, the proposed control structure is very simple and is more easily implemented than those of conventional control schemes.

Motivated by the aforementioned exploration, the major contributions of this study are as follows:

- A PD-type attitude control is proposed so that desirable performance for the attitude quaternion and rotation velocity in spite of the actuators saturation are provided.
- The performance criteria such as overshoot, convergence rate and ultimate value of the system states do not depend on the control parameters and they can be defined by the finite-time prescribed performance function (FTPPF) which will be given later.
- Compared to the constrained controls based on the PPC notion consisting of intricate terms [30]–[36], the proposed constrained PD-type controller has a simple structure that is of considerable importance in practice.

The rest of this study is arranged as follows. In Section 2, the rigid body nonlinear model and attitude control problem are stated. The main result is presented in Section 3, where a simple structure constrained PD-type control is given to acquire high accurate attitude control. Eventually, Sections 4 and 5, respectively, contain numerical simulations and conclusions.

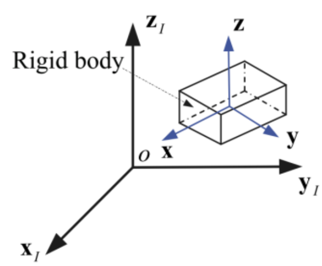

FIGURE 1. The orientation description of a rigid body [40].

II. PROBLEM FORMULATION

A. MATHEMATICAL MODEL OF A RIGID BODY

In order to provide the rigid body attitude dynamics, two coordinate frames namely the inertial-fixed frame $F_i(x_I - y_I - z_I)$ and the body-fixed frame $F_b(x - y - z)$ are utilized (see FIGURE 1). This paper employs the unit quaternion for representing the rigid body attitude in F_b with respect to (w.r.t) F_i. The dynamical model for a rigid body attitude system can be expressed by the following differential equations [39]

$$\begin{aligned}\dot{\boldsymbol{q}}_v &= \frac{1}{2}(q_4\boldsymbol{I}_3 + \boldsymbol{q}_v^{\times})\boldsymbol{\omega}; \quad \dot{q}_4 = -\frac{1}{2}\boldsymbol{q}_v^T\boldsymbol{\omega}\\ \boldsymbol{J}\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega}^{\times}\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{u} + \boldsymbol{d}\end{aligned} \tag{1}$$

in which the unit quaternion $\boldsymbol{q} = [\boldsymbol{q}_v^T,\ q_4]^T = [q_1,\ q_2,\ q_3,\ q_4]^T$ is used for the rigid body attitude representation, where $\boldsymbol{q}^T\boldsymbol{q} = 1$. Moreover, $\boldsymbol{\omega} \in R^3 = [\omega_1,\ \omega_2,\ \omega_3]^T$ is the rotation velocity of F_b w.r.t F_i and expressed in F_b, $\boldsymbol{J} \in R^{3\times3}$ is the positive-definite inertia matrix of the rigid body, $\boldsymbol{u} = [u_1,\ u_2,\ u_3]^T \in R^3$ is the control torque, and $\boldsymbol{d} = [d_1,\ d_2,\ d_3]^T \in R^3$ denotes the unknown but bounded disturbances where $\|\boldsymbol{d}\| \le \bar{d}$. For any given vector $\boldsymbol{z} = [z_1,\ z_2,\ z_3]^T \in R^3$, then $\boldsymbol{z}^{\times}$ is a cross-product matrix defined as

$$\boldsymbol{z}^{\times} = \begin{bmatrix} 0 & -z_3 & z_2\\ z_3 & 0 & -z_1\\ -z_2 & z_1 & 0\end{bmatrix} \tag{2}$$

Let $\boldsymbol{sat}(\boldsymbol{u})$ be the vector of actual control input produced by the actuators. The saturation function is expressed as

$$sat(u_i) = u_i(t) + \pi_i(t) \tag{3}$$

where

$$\pi_i(t) = \begin{cases} 0; & |u_i(t)| < u_{\max i}\\ \mathrm{sgn}(u_i(t))u_{\max i} - u_i(t); & |u_i(t)| \ge u_{\max i}\end{cases} \tag{4}$$

where $u_{\max i}$ is the maximum control torque generated by the actuators. Note that $\boldsymbol{\Pi}(t) = [\pi_1(t)\pi_2(t)\pi_3(t)]^T$ is unknown but bounded.

Thus, the attitude system dynamics in the presence of the actuator saturation can be rewritten as

$$\begin{aligned}\dot{\boldsymbol{q}}_v &= \frac{1}{2}(q_4\boldsymbol{I}_3 + \boldsymbol{q}_v^{\times})\boldsymbol{\omega}; \quad \dot{q}_4 = -\frac{1}{2}\boldsymbol{q}_v^T\boldsymbol{\omega}\\ \boldsymbol{J}\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega}^{\times}\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{u} + \boldsymbol{T}_d\end{aligned} \tag{5}$$

where $\boldsymbol{T}_d = \boldsymbol{d} + \boldsymbol{\Pi}$ and $\|\boldsymbol{T}_d\| \le \boldsymbol{\theta}$ such that θ is a positive unknown constant.

B. PROBLEM FORMULATION

1) PROBLEM 1

Consider the rigid body attitude system (5) subject to external disturbance and the actuators saturation. The control objective is to develop a simple structure PD-type attitude control such that:

- The rigid body attitude maneuver is accomplished and desirable performance for the quaternion and the angular velocity are provided, simultaneously.
- The aforementioned objectives are satisfied under the actuator physical limitation.

III. MAIN RESULTS

To solve Problem 1, a novel FTPPF and a simple structure constrained PD-type approach are developed.

A. PRESCRIBED PERFORMANCE FUNCTION

Based on the concept of PPC [22], to satisfy the constraint on the attitude variable, the first step is to define a PPF specifying the performance in transient and steady state including maximum overshoot, convergence rate and ultimate attitude value. In [22], it has been rigorously explained that the predefined performance for the attitude variable is obtained if it is remained within a specific area which is bounded by the PPF. In accordance with this notion, to provide any desirable performance for the attitude variable, it is adequate to ensure that

$$-\rho_q(t) < q_i(t) < \rho_q(t), \tag{6}$$

where the PPF $\rho_q(t)$ is usually defined as an exponentially decaying function as follows

$$\rho_q(t) = \left(\rho_{q0} - \rho_{q\infty}\right)\exp(-\kappa_q t) + \rho_{q\infty}, \tag{7}$$

where ρ_{q0}, $\rho_{q\infty}$ and κ_q are positive design parameters that ought to be suitably chosen based on the requirements. Each of these parameters is selected to provide a specification for the time response of the quaternion. To be more exact, ρ_{q0} determines the maximum overshoot, κ_q specifies the convergence speed of the quaternion trajectory and $\rho_{q\infty}$ characterizes the ultimate bound on the quaternion.

According to the definition of PPF $\rho_q(t)$ in (7), it reaches its ultimate value $\rho_{q\infty}$ as time tends to infinity. In contrast to the existing PPFs with exponential convergence, we introduce a finite-time PPF (FTPPF) as:

$$\rho(t) = \begin{cases}\left(\rho_0 - \rho_T(1 + t/T_s)\right)\exp(\kappa\, t/(t - T_s)) + \rho_T; & 0 \le t < T_s\\ \rho_T; & t \ge T_s\end{cases} \tag{8}$$

in which $\rho_T = \lim\limits_{t\to T_s}\rho(t)$ and T_s denotes the settling time of the FTPPF.

Remark 1: Note that the fundamental difference between PPF (6) and FTPPF (8) lies in the fact that that the latter possesses finite-time convergence behavior. As can be seen in FIGURE 2, both of them have the same ρ_0 and ρ_T. However, the FTPPF (8) arrives ρ_T prior to $T_s = 6$ sec and stays thereafter whereas PPF (6) has not reached yet.

B. PRESCRIBED PERFORMANCE CONTROL

To accomplish the control objective stated in Problem 1, it is sufficient to ensure the following relationships are satisfied

$$\begin{aligned}-\rho_q(t) &< q_i(t) < \rho_q(t),\\ -\rho_\omega(t) &< \omega_i(t) < \rho_\omega(t),\end{aligned} \tag{9}$$

where $i = 1, 2, 3$. Based on the condition (9), it is concluded that if $|q_i(0)| < \rho_{q0}$ and $|\omega_i(0)| < \rho_{\omega0}$ hold, the attitude

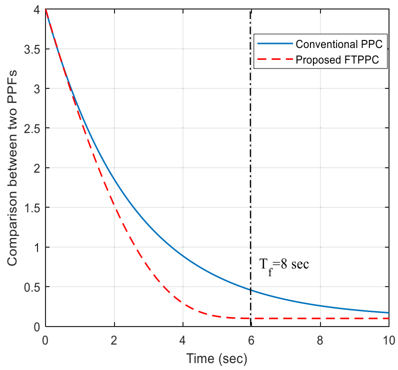

FIGURE 2. Comparison between PPF (6) and FTPPF (8).

quaternion and rotation velocity always remain in the predefined bounds. Applying the PPC concept [32]–[36], [41], two transformed variables for the quaternion and angular velocity should be adopted to transform the original nonlinear attitude system (5), into an equivalent unconstrained one. More specifically, it is defined

$$\begin{aligned} q(t) &= \rho_q(t)\, S_q(\varepsilon_q) \\ \omega(t) &= \rho_\omega(t)\, S_\omega(\varepsilon_\omega) \end{aligned} \tag{10}$$

where ε_q and ε_ω are the transformed variables and $S_q(\varepsilon_q)$ and $S_\omega(\varepsilon_\omega)$ are smooth, strictly increasing and invertible functions, for instance $S(\varepsilon(t)) = \frac{e^{\varepsilon(t)} - e^{-\varepsilon(t)}}{e^{\varepsilon(t)} + e^{-\varepsilon(t)}}$.

Because of the properties of $S(\cdot)$ as well as $\rho_q(t) > 0$ and $\rho_\omega(t) > 0$, the inverse transformation $\varepsilon_q = S_q^{-1}\left(q(t)/\rho_q(t)\right)$ and $\varepsilon_\omega = S_\omega^{-1}\left(\omega(t)/\rho_\omega(t)\right)$ are well defined if (9) holds or equivalently $\varepsilon_q,\ \varepsilon_\omega \in L_\infty$. Thus, if ε_q and ε_ω are guaranteed to be kept bounded, then (9) is ensured. It should be noted that to stabilize the new transformed variables associated with the quaternion and angular velocity, their dynamics should be obtained. Thus, we will face a highly nonlinear with strong coupling system for which designing a controller will give rise to severe difficulties. However, in what follows, we present a straightforward way to constrain the quaternion and angular velocity, simultaneously.

C. CONSTRAINED PD-TYPE CONTROL

By virtue of the concept of PPC concept, a new methodology is proposed in this section to deal with the difficulty arisen from stabilizing the transformed errors associated with the quaternion and angular velocity. Let us define the following simple error transformations

$$\boldsymbol{\varepsilon}_1 = \frac{1}{\rho_q(t)}\boldsymbol{q}_v(t), \quad \boldsymbol{\varepsilon}_2 = \frac{1}{\rho_\omega(t)}\boldsymbol{\omega}(t) \tag{11}$$

where $\boldsymbol{\varepsilon}_1 = [\varepsilon_{11}, \varepsilon_{12}, \varepsilon_{13}]^T \in R^3$ and $\boldsymbol{\varepsilon}_2 = [\varepsilon_{21}, \varepsilon_{22}, \varepsilon_{23}]^T \in R^3$. In light of (11), if $|\varepsilon_{li}(t)| < 1$ for $i = 1, 2, 3$, $l = 1, 2$ holds, then the performance constraints on the quaternion and angular velocity are satisfied. Consequently, the main control objective becomes to design a controller which is able to keep the absolute value of $\varepsilon_{li}(t)$ less than 1. To this end, the following simple structure constrained PD-type attitude control is proposed.

$$\boldsymbol{u} = -k_p \boldsymbol{q}_v - k_d \boldsymbol{\omega} - \boldsymbol{u}_{cons} \tag{12}$$

where k_p and k_d are positive scalars and

$$\boldsymbol{u}_{cons} = \left[\alpha(1-\boldsymbol{\varepsilon}_1)^{-2} + \beta(1-\boldsymbol{\varepsilon}_2)^{-2}\right]\boldsymbol{s} + \hat{\theta}\,\mathrm{sgn}(\boldsymbol{s}) \tag{13}$$

where α and β are positive constants, $\boldsymbol{s} = \boldsymbol{\omega} + c\boldsymbol{q}_v$ with $c > 0$ and $\hat{\theta}$ is the estimation of θ and it is updated by the following mechanism

$$\dot{\hat{\theta}} = \frac{1}{\eta_1}\left(\|\boldsymbol{s}\| - \eta_2\hat{\theta}\right). \tag{14}$$

In adaptive law (14), $\eta_1 > 0$ and η_2 is a positive function of time such that $\int_0^\infty \eta_2(\tau)d\tau = \bar{\eta} < \infty$.

Theorem 1: Consider the attitude system (5). Applying the constrained PD-type attitude control (12), then the system trajectories converge to the origin as time tends to infinity.

Proof: Construct the following Lyapunov function which is slightly different from that of [42]

$$\begin{aligned} V = (k_p + ck_d)\left(\boldsymbol{q}_v^T\boldsymbol{q}_v + (1-q_0^2)\right) + \frac{1}{2}\boldsymbol{\omega}^T\boldsymbol{J}\boldsymbol{\omega} \\ + c\boldsymbol{q}_v^T\boldsymbol{J}\boldsymbol{\omega} + \frac{1}{\eta_1}\tilde{\theta}^2 \end{aligned} \tag{15}$$

where $\tilde{\theta} = \theta - \hat{\theta}$. The Lyapunov function (15) is bounded by

$$V \geq \frac{1}{2}\boldsymbol{x}^T\boldsymbol{P}\,\boldsymbol{x} \tag{16}$$

where $\boldsymbol{x} = \left[\|\boldsymbol{q}_v\|, \|\boldsymbol{\omega}\|, \tilde{\theta}\right]^T$, $\boldsymbol{P} = \frac{1}{2}\begin{bmatrix} 2(k_p+ck_d) & -c\lambda_{\max}(\boldsymbol{J}) & 0 \\ -c\lambda_{\max}(\boldsymbol{J}) & \lambda_{\min}(\boldsymbol{J}) & 0 \\ 0 & 0 & 1/2\eta_1 \end{bmatrix}$, and $\lambda_{\max}(\boldsymbol{J})$ and $\lambda_{\min}(\boldsymbol{J})$ refer to the maximum and minimum eigenvalue of the inertia matrix, respectively. For c sufficiently small, the matrix $\boldsymbol{P}$ is positive definite. The time-derivative of the Lyapunov function along (5) and substituting the control law (12) give

$$\begin{aligned} \dot{V} &= -ck_p\left\|\boldsymbol{q}_v\right\|^2 - k_d\left\|\boldsymbol{\omega}\right\|^2 + \boldsymbol{s}^T\boldsymbol{T}_d + \frac{c}{2}\left(q_4\boldsymbol{\omega} + \boldsymbol{\omega}^\times\boldsymbol{q}_v\right)\boldsymbol{J}\boldsymbol{\omega} \\ &\quad - \boldsymbol{s}^T\left[\alpha(1-\boldsymbol{\varepsilon}_1)^{-2} + \beta(1-\boldsymbol{\varepsilon}_2)^{-2}\right]\boldsymbol{s} \\ &\quad - \|\boldsymbol{s}\|\,\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}\left(\|\boldsymbol{s}\| - \eta_2\hat{\boldsymbol{\theta}}\right) \end{aligned} \tag{17}$$

Note that for any positive constant σ, the following relation is satisfied

$$\tilde{\theta}\hat{\theta} \leq -\frac{2\sigma - 1}{2\sigma}\tilde{\theta}^2 + \frac{\sigma}{2}\theta^2 \tag{18}$$

Then, the inequality (17) can be simplified as

$$\dot{V} \leq -\boldsymbol{x}^T\boldsymbol{Q}\,\boldsymbol{x} + \frac{\eta_2\sigma}{2}\theta^2 \leq -\lambda_{\min}(\boldsymbol{Q})\,\|\boldsymbol{x}\|^2 + \frac{\eta_2\sigma}{2}\theta^2 \tag{19}$$

where $\boldsymbol{Q} = \begin{bmatrix} ck_p & 0 & 0 \\ 0 & k_d - c\lambda_{\max}(\boldsymbol{J}) & 0 \\ 0 & 0 & \eta_2(2\sigma-1)/2\sigma \end{bmatrix}$. For c sufficiently small, the matrix $\boldsymbol{Q}$ is positive definite, i.e.

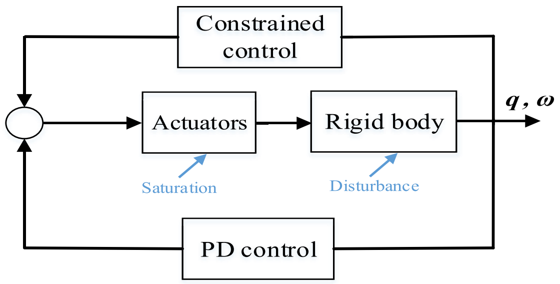

FIGURE 3. Structure of the proposed attitude control system.

$\lambda_{\min}(\boldsymbol{Q}) > 0$. Integrating both sides of (19) results in

$$V(t) \leq V(0) - \lambda_{\min}(\boldsymbol{Q}) \int_0^t \|\boldsymbol{x}(\tau)\|^2 d\tau + \frac{\sigma}{2}\theta^2 \int_0^t \eta_2(\tau) d\tau \tag{20}$$

Thus, the boundedness of $V(t)$ can be inferred. Then, the quaternion and rotation velocity and their derivatives are bounded. In light of (20), one obtains

$$\|\boldsymbol{x}(\tau)\|_{L_2}^2 \leq \frac{\bar{\eta} + V(0)}{\lambda_{\min}(\boldsymbol{Q})} \tag{21}$$

Therefore, $\boldsymbol{q}_v$, $\boldsymbol{\omega} \in L_2$. Applying the Barbalat's lemma [43], it can be inferred that $\lim_{t\to\infty} \boldsymbol{q}_v = 0$ and $\lim_{t\to\infty} \boldsymbol{\omega} = 0$. The proof is finished here. ■

Remark 2: The function $(1-\boldsymbol{\varepsilon}_1)^{-2}$ is always positive. Thus, $-\boldsymbol{s}^T(1-\boldsymbol{\varepsilon}_1)^{-2}\boldsymbol{s} < 0$ is ensured and it can be removed from the left hand side of the inequality (17). It should also be pointed out that when the quaternion approaches its boundary ρ_q, then ε_{1i} tends to one and function $(1-\boldsymbol{\varepsilon}_1)^{-2}$ increases to prevent the quaternion from growing and steers it to zero. The similar explanation is true for the angular velocity.

Remark 3: As can be observed in (12), the new constrained control possesses much simpler structure in comparison with the existing constrained attitude controls. In fact, the suggested controller contains two parts: the first one is a conventional PD control whose task is to drive the system states to zero and the second one is an auxiliary control. The second term of the final controller is employed to satisfy the constraints on the quaternion and angular velocity and to compensate for the external environmental disturbance and actuator saturation. Thus, if the proposed control is applied, the desired performance in transient and steady state such as overshoot, convergence time and ultimate bound of the system states are guaranteed.

Remark 4: The structure of the proposed attitude control system is illustrated in FIGURE 3 to give a deeper understanding of the overall approach.

Remark 5: Since discontinuous controllers suffer from chattering, one way to alleviate this problem is to consider an approximation of the signum function by a saturation function with a high slope $1/\varsigma$. Thus, the controller (12) is updated as

$$\boldsymbol{u} = -k_p \boldsymbol{q}_v - k_d \omega - \left[\alpha(1-\varepsilon_1)^{-2} + \beta(1-\varepsilon_2)^{-2}\right] \boldsymbol{s} - \hat{\theta} sat(\boldsymbol{s}/\varsigma) \tag{22}$$

Remark 6: As can be seen in the controller (12), the parameters k_p and k_d are related to the PD control. Thus, the larger k_p is selected, the faster convergence and smaller steady-state error can be obtained. However, it can increase the overshoot and also control input. The gain k_d will have the effect of increasing the stability of the system, reducing the overshoot, and improving the transient response. Nevertheless, it can increase the settling time. The parameters α and β weight the distance between the quaternion or angular velocity trajectory with their corresponding PPF. For instance, when α becomes larger, the distance between the trajectory of the quaternion and its corresponding PPF is kept larger. Moreover, in the FTPPF (8), ρ_0, ρ_∞, and T_s are positive design parameters that ought to be suitably chosen based on the requirements. Each of these parameters is selected to provide a specification for the time response of the quaternion. To be more exact, ρ_0 determines the maximum overshoot, T_s shows the settling time and ρ_∞ characterizes the ultimate bound.

Remark 7: It is worth mentioning that the condition (9) is equivalent to $|\varepsilon_{1i}(t)| < 1$ and $|\varepsilon_{2i}(t)| < 1$. Thus, based on (9) and (11), it is concluded that if $|q_i(0)| < \rho_{q0}$ and $|\omega_i(0)| < \rho_{\omega 0}$ hold, then $|\varepsilon_{1i}(0)| < 1$ and $|\varepsilon_{2i}(0)| < 1$. When the attitude quaternion or angular velocity increases, then it tends to its corresponding PPF, i.e., $|q_i(t)| \to \rho_q(t)$ or $|\omega_i(t)| \to \rho_\omega(t)$. This makes $\boldsymbol{\varepsilon}_1 \to \boldsymbol{I}$ or $\boldsymbol{\varepsilon}_2 \to \boldsymbol{I}$, and the value of $(1-\boldsymbol{\varepsilon}_1)^{-2}$ or $(1-\boldsymbol{\varepsilon}_2)^{-2}$ increases. At the same time, the controller counteracts the attitude quaternion and angular velocity to suppress its evolvement.

Remark 8: To implement the proposed scheme, the following algorithm is given.

Algorithm 1 The implementation of the proposed algorithm

Initialization(off-line):
Calculate the control gains:
1: Select appropriate $\eta_1 > 0$ and η_2 without considering given performance;
2: Choose the weighting gains α and β;
3: Select appropriate values for the PD control part;
4: Choose the parameters of the FTPPF (8) based on the predefined overshoot, convergence rate and steady-state error;
Initial states:
1: Initialize $\boldsymbol{q}(0)$ and $\boldsymbol{\omega}(0)$;
2: Initialize the adaptive estimator $\hat{\theta}(0)$;
3: Set the input saturation upper bound $u_{\max i}$;
At sampling time $t = t_k$ (on-line):
1: Update the state signals $\boldsymbol{q}$ and $\boldsymbol{\omega}$;
2: Calculate $\boldsymbol{\varepsilon}_1$ and $\boldsymbol{\varepsilon}_2$ from (11);
3: Compute the designed controller $\boldsymbol{u}$ and substitute it in the dynamic (5);
4: Update the adaptive estimator $\hat{\theta}(t_{k+1})$ from (14);
At sampling time $t = t_{k+1}$ (on-line):
Repeat the steps at time $t = t_{k+1}$.

IV. SIMULATION RESULTS

This section aims to evaluate the usefulness and efficacy of the suggested control framework for accurate attitude control. To this end, the constrained PD-type control (12) is applied to a rigid spacecraft and the results are given in two parts. The inertia matrix of the considered spacecraft is given by $\boldsymbol{J} = [20\ 1.2\ 0.9;\ 1.2\ 17\ 1.4;\ 0.9\ 1.4\ 15]\, kg.m^2$ [44]. The external environmental disturbances is given as $d(t) = [0.05\sin(t),\ 0.1\ \sin(1.2t),\ 0.15\sin(1.5t)]^T\ Nm$. It is supposed that the actuators are not able to produce torques more than 2.5 Nm. The parameters of the constrained PD-type control are selected as $K_p = 40$, $K_d = 20$, $\alpha = 1.5$, $\beta = 1.2$, $\eta_1 = 0.1$, $\eta_2 = 2\exp(-0.5t)$ and $\hat{\theta}(0) = 0$. The parameters of the FTPPFs ρ_q and ρ_ω are given as $\rho_{q0} = 0.3$, $\rho_{\omega 0} = 0.4$, $\rho_{q\infty} = 0.008$, $\rho_{\omega\infty} = 0.005$, $\sigma_q = 2.5$, and $\sigma_\omega = 2.5$.

A. PART 1

We are about to show that how changing the parameters of the FTPPF (8) especially its settling time can affect the convergence rate of the system trajectories. For this purpose, the simulation is run for two cases in which T_f is 15 and 10 sec. The initial attitude and rotation velocity of the spacecraft are given as $\boldsymbol{q}(0) = [0.2\ -0.15\ -0.25\ 0.9354]^{\mathrm{T}}$ and $\boldsymbol{\omega}(0) = [0.03,\ 0.02,\ -0.01]^{\mathrm{T}}\ \mathrm{rad}/\mathrm{sec}$.

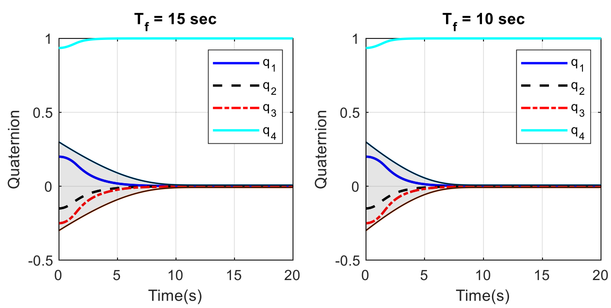

FIGURE 4. The time response of the quaternions in part 1.

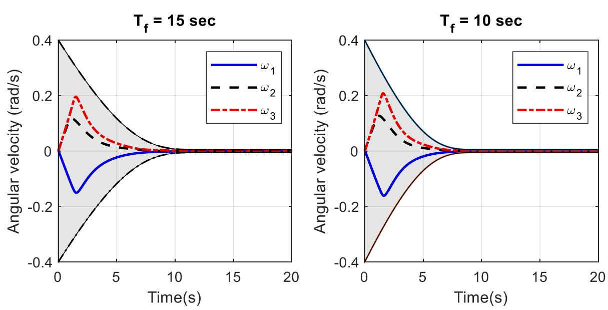

FIGURE 5. The time response of the angular velocity in part 1.

According to FIGURE 4 and FIGURE 5, the trajectories of the attitude quaternion and rotation velocity are kept within the allowable regions confined between $-\rho(t)$ and $\rho(t)$ which is highlighted in gray color. As expected, the controller does not allow the trajectories contact the boundaries and violate the constraints. Thus, the favorable performance specifications in transient and steady state are obtained.

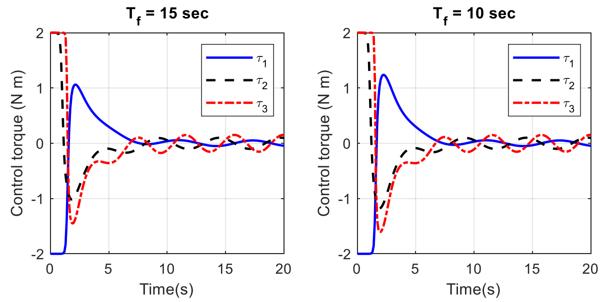

FIGURE 6. The time response of the control torque in part 1.

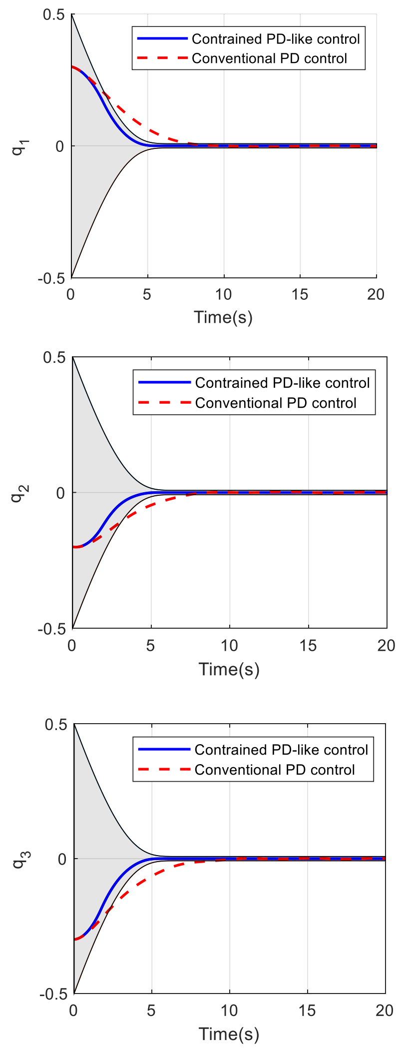

FIGURE 7. The time response of the quaternions in part 2.

Besides, the smaller T_f is, the faster convergence rate for the system trajectories is obtained. FIGURE 6 shows that the maximum required control torque is within the allowable limit.

B. PART 2

In this part, the conventional PD control as a widely used control strategy in spacecraft attitude control system is selected to make a comparison with the proposed control

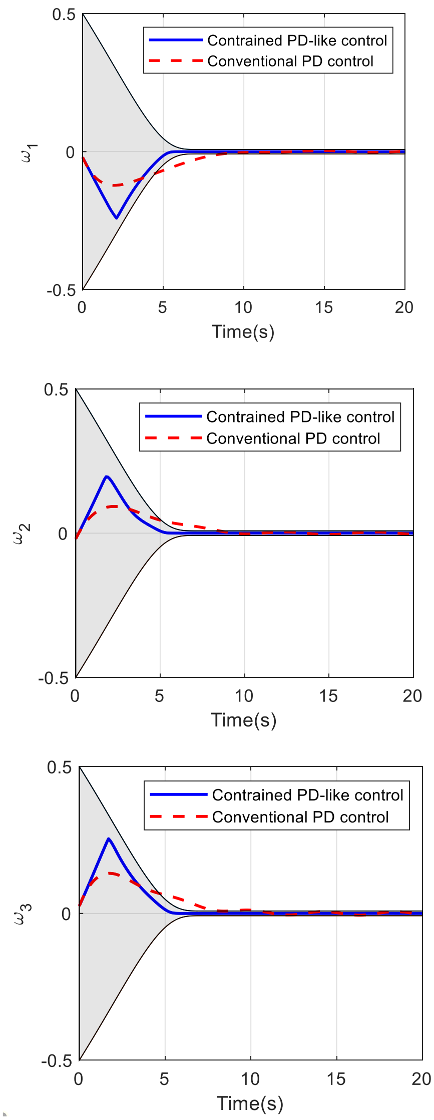

FIGURE 8. The time response of the angular velocity in part 2.

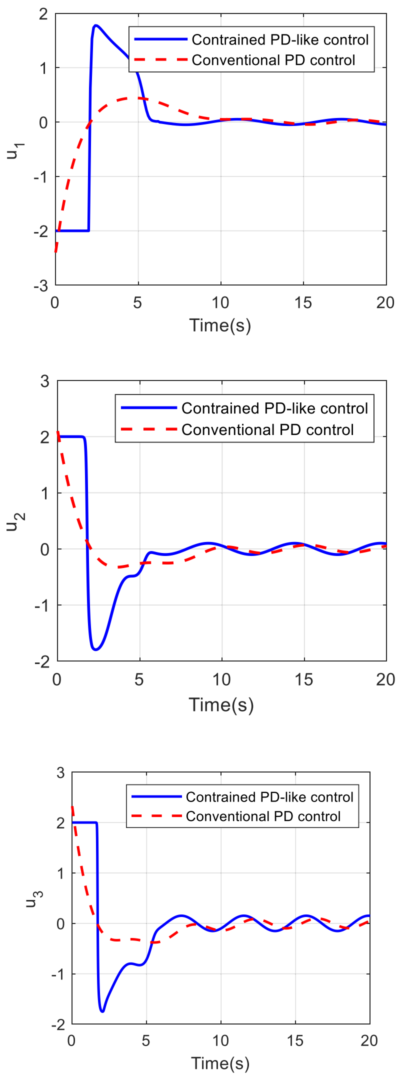

FIGURE 9. The time response of the control torque in part 2.

approach. The initial conditions for attitude quaternion and rotation velocity are $\boldsymbol{q}(0) = [0.3 \ \ -0.2 \ \ -0.3 \ \ 0.8832]^{\mathrm{T}}$ and $\boldsymbol{\omega}(0) = [-0.02, \ -0.02, \ 0.025]^T$ rad/sec. The simulation results related to this part are provided in FIGURE 7 to FIGURE 9.

FIGURE 7 and FIGURE 8 are respectively illustrate the time responses of the vector part of the attitude quaternion and rotation velocity under the constrained PD-type control (12) and conventional PD control. As can be observed, using the suggested controller, the attitude and angular velocity trajectories are confined within the highlighted predefined regions for them and the constraints are satisfied. However, the PD control is not capable of providing as fast and accurate convergence as the proposed control. These results are obtained under the situation that both methods require the same maximum control effort as it is seen in FIGURE 9. Thus, with similar maximum control torque, the constrained PD-type control achieves much better performance.

V. CONCLUSION

A constrained PD-type attitude control was presented for a rigid body with guaranteed desired performance of the attitude quaternion and the rotation velocity. The advantages of the proposed control framework are twofold: 1) Unlike the existing constrained controls in the literature, the novel PD-type control has a simple structure and provides specific transient and steady-state performance for the attitude quaternion and the rotation velocity of the rigid body. 2) A FTPPF was introduced to assure that the system trajectories converge to their ultimate regions within a finite time independent of the initial conditions. 3) A rigorous proof is given to show that the closed-loop attitude system is stable in the sense of Lyapunov stability concept. The efficiency of the suggested approach was assessed using numerical simulations carried out on a rigid spacecraft and compared to that of the conventional PD attitude control. The obtained results confirmed the highly desirable performance of the suggested control scheme.

It should be pointed out that for very rapid maneuvers, it is necessary to consider the actuator saturation and rate, simultaneously. Designing a constrained control subject to actuator saturation and rate is one of our future works.

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